

MODULE 3

QUANTUM MECHANICS

Up to 19th century, the old classical mechanics proposed by Newton was enough to explain all motion of bodies . In 1900 Max plank explained black body spectrum .In order to understand the world of atomic or subatomic scale, a new branch in physics is necessary. Quantum mechanics is the science of microscopic world

COMPARISON BETWEEN NEWTONIAN MECHANICS & QUANTUM MECHANICS

NEWTONIAN MECHANICS

1. Related to macroscopic universe
2. Deals with motion under applied forces
3. Predicted values agrees with the observed values
4. Gives the correct behavior of moving bodies
5. Based on Newton's laws

QUANTUM MECHANICS

1. Related to microscopic universe
2. Based on uncertainty principle
3. Gives the relation between predicted value and observed value
4. Gives the probable value.
5. Schrodinger equation is the fundamental of Q.M

MATTER WAVES OR DE BROGLIE WAVES

Louis de Broglie introduced the concepts of matter waves. De Broglie argued that nature was symmetrical, then the fundamental quantities matter and energy should display symmetrical characters. If light shows dual nature then matter also shows particle nature and wave nature.

De Broglie hypothesis says that every moving matter exhibit wave like properties. The wave associated with a moving particle is called matter wave .

DE BROGLIE WAVELENGTH

Let c be the velocity of light. consider a photon of momentum p and frequency ν

$$mc^2 = h\nu$$

The momentum of the photon, $p = mc = h\nu/c$

$$\frac{\nu}{c} = \frac{1}{\lambda}$$

$$p = \frac{h}{\lambda} \quad \text{ie} \quad \lambda = \frac{h}{p}$$

If m is the mass and v is the velocity of the particle, then

De Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

This is known as de Broglie wave equation.

DE BROGLIE WAVELENGTH FOR ELECTRONS

Consider an electron of mass m and charge e subjected to a potential difference of V volts. If v is the velocity acquired by the electron, then

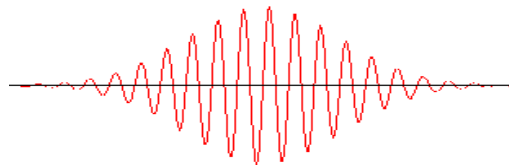
$$\frac{1}{2}mv^2 = eV$$

then wavelength

$$\lambda = \frac{h}{\sqrt{2meV}} = \sqrt{\frac{150}{V}} \text{Å} = \frac{12.3}{\sqrt{V}} \text{Å}$$

The wavelength of electron is inversely proportional to the square root of accelerating potential.

A WAVE PACKET



According to De Broglie hypothesis a wave is associated with a moving particle . Hence a particle can be represented by a wave confined in space .

A Wave that is confined to a small region of space in the vicinity of the particle is called a wave packet. It is the envelope of number of wave superimposed. In Quantum mechanics a *particle can be represented by a wave packet*. They are not electromagnetic waves. The velocity of matter waves is not constant and it is always greater than the velocity of light

WAVE PACKET, PHASE VELOCITY AND GROUP VELOCITY

The velocities of the individual waves which superimpose to produce the wave packet representing the particle are different - the wave packet as a whole has a different velocity from the waves that comprise it

Phase velocity, v : The velocity of individual waves that forming a wave packet.

Group velocity, u : The velocity with which wave packet moves .

$$u = \frac{c^2}{v}$$

WAVE FUNCTION

$$\Psi(r, t)$$

- I. Gives character about matter waves.
- II. Gives the probability of finding the particle at particular point and time

PHYSICAL CONCEPT (significance) OF WAVE FUNCTION

The quantity whose variation makes up the matter waves is called the wave function Ψ . The quantity that undergoes periodic changes of a body is Ψ .

$$\Psi = Ae^{i(kx - \omega t)}$$

Ψ is a complex valued function and itself has no physical meaning. The square of the absolute magnitude/ $|\Psi|^2$ or $\Psi \Psi^* dx dy dz$ gives the probability of finding the particle in the volume element $dx dy dz$. We can obtain the physical properties of the system if we know the wave function.

Since $\Psi \Psi^* dx dy dz$ is proportional to the probability of finding the particle with in the volume element, the integral $\int \Psi \Psi^* dx dy dz$ must be finite if the particle is somewhere there. If $\int \Psi \Psi^* dx dy dz$ is zero, the particle doesn't exist and if it is infinity, the particle is everywhere simultaneously.

Since the probability of finding the particle in the volume element is a surety, then

$$\int \Psi \Psi^* dx dy dz = 1$$

The wave function which obeys this condition is called normalized wave function.

REQUIREMENTS OF WAVE FUNCTION

1. Ψ must be finite, single valued and continuous.
2. $\partial \Psi / \partial x$, $\partial \Psi / \partial y$, $\partial \Psi / \partial z$ must also be continuous and single valued everywhere.
3. Ψ can be normalized

HEISENBERG'S UNCERTAINTY PRINCIPLE

According to the classical mechanics the position and momentum of a particle can be determined simultaneously with accuracy. In Newtonian mechanics every particle has a fixed position in space and has a definite momentum at any time. But the concept of dual nature, particle and wave, inspired Heisenberg to state the principle of uncertainty.

According to this principle it is impossible to have an accurate measurement of the position and momentum of particles simultaneously.

or

The Uncertainty Principle states that the simultaneous determination of the exact position and momentum of a moving particle is impossible

The product of uncertainty (error) in the measurement of position of the particle (Δx) and the uncertainty in the momentum (ΔP_x) of the particle is of the order of Planks constant.

$$\Delta x \cdot \Delta P_x = \hbar \quad (\hbar = h/2\pi)$$

$$\Delta y \cdot \Delta P_y = \hbar$$

$$\Delta z \cdot \Delta P_z = \hbar$$

If Δx is small, ΔP_x will be large and vice versa. If one quantity is measured accurately, the other quantity becomes less accurate. If $\Delta x = 0$, $\Delta P_x = \text{infinity}$; that is if we know the exact position of a particle we shall never know its momentum with certainty. If $\Delta P_x = 0$, $\Delta x = \text{infinity}$; that is if we know the momentum of particle, its position is not known.

Position & momentum

$$\Delta x \Delta p_x = \frac{h}{2\pi}$$

Similarly in the simultaneous measurement of energy and time, if ΔE and Δt are the uncertainties we have $\Delta E \Delta t = \hbar$

Energy & time

$$\Delta E \Delta t = \frac{h}{2\pi}$$

Angular position and angular momentum

$$\Delta J \Delta \theta = \frac{h}{2\pi}$$

APPLICATIONS OF UNCERTAINTY PRINCIPLE

1. Uncertainty in frequency of light emitted by an atom

An electron exists in an excited state only for a short interval of time. Thus Δt is small, then ΔE must be large

$$\Delta t = 10^{-8} \text{ s}$$

$$\Delta E \Delta t = \hbar$$

The uncertainty in energy, $\Delta E = \hbar / 10^{-8}$

$$E = h \nu$$

$$\Delta E = h \Delta \nu$$

$$\Delta \nu = \Delta E / h$$

$$\Delta \nu = \left[\frac{\frac{h}{2\pi}}{10^{-8}} \right] * \frac{1}{h} = 1.67 * 10^7 \text{ Hz}$$

2. Non existence of electron in the nucleus

The nuclear diameter is of the order of 10^{-15} m . If an electron exists in the nucleus, the maximum uncertainty, Δx , in position is of the order of the diameter.

So the uncertainty in the momentum,

$$\begin{aligned} \Delta p &= \frac{\hbar}{\Delta x} = \frac{6.626 \times 10^{-34}}{2 \times 3.14 \times 10^{-15}} \\ &= 1.055 \times 10^{-19} \text{ kg m/s} \end{aligned}$$

The energy of electron is given by

$$\begin{aligned}
 E &= pc = 1.055 \times 10^{-19} \times 3 \times 10^8 \\
 &= 3.165 \times 10^{-11} \text{ J} \\
 &= 3.165 \times 10^{-11} / 1.6 \times 10^{-19} \text{ eV} \\
 &= 1.98 \times 10^8 \text{ eV} \\
 &= 198 \text{ MeV}
 \end{aligned}$$

For an electron to exist in the nucleus, it should have an energy of this order. Energy of an electron is of the order of only a few MeV. So it is not possible for electron to exist inside nucleus

3. Natural line broadening

The average time period that takes between the excitation of an atom and the time it radiates is 10^{-8} . Thus the photon energy is uncertain by the amount

$$\begin{aligned}
 \Delta E &= \hbar / \Delta t \\
 &= 1.055 \times 10^{-26} \text{ J}
 \end{aligned}$$

It means that the excited energy levels have a finite energy spread. Thus the energy levels of the atom must have a finite width or in other words the spectral lines can never be sharp but have some broadening in frequency, called natural line broadening.

OPERATORS

An operator transforms one function to another function. If the operator A transforms $f(x)$ into the function $g(x)$, then we can write

$$g(x) = A f(x)$$

In Q.M each dynamic variable is represented by an operator which acts on a wave function to give a new wave function.

The operators associated with some of the dynamical variables is given below

Dynamical Variable	Quantum Mechanical operator
time t	t
Position Co-ordinates x, y, z	x, y, z
Momentum co-ordinates p_x, p_y, p_z	$-i\hbar \frac{\partial}{\partial x}, -i\hbar \frac{\partial}{\partial y}, -i\hbar \frac{\partial}{\partial z}$
Momentum p	$-i\hbar \nabla$
Energy E	$i\hbar \frac{\partial}{\partial t}$

TIME DEPENDENT SCHRÖDINGER EQUATION FOR A FREE PARTICLE

Schrodinger's equation is the basic expression in quantum mechanics. We can derive it by considering the plane wave equation and combining with Einstein's eqn for quantum energy and de Broglie expression for wavelength.

A particle in motion is associated with a wave function that contains the information about the motion.

A plane wave that propagates along x- direction is given by

$$\Psi = Ae^{i(kx - \omega t)}$$

where k is the wave vector, $k = 2\pi/\lambda$

ω is the angular frequency, $\omega = 2\pi\nu$

Einstein's formula for photon energy is

$$E = h\nu = \frac{h\omega}{2\pi} = \hbar\omega; \text{ where } \hbar = \frac{h}{2\pi}$$

de-Broglie's expression for matter wavelength is $\lambda = \frac{h}{p}$

$$\therefore p = \frac{h}{\lambda} = \frac{h}{2\pi} \cdot \frac{2\pi}{\lambda} = \hbar k$$

Using the above expression, equation (1) becomes

$$\Psi = Ae^{\frac{i}{\hbar}(\hbar kx - \hbar\omega t)}$$

$$\text{ie } \Psi = Ae^{\frac{i}{\hbar}(px - Et)}$$

On partial differentiation of Ψ with respect to x, twice, we get

$$\frac{\partial^2 \Psi}{\partial x^2} = \frac{-p^2}{\hbar^2} \Psi \dots\dots\dots(2) \quad \text{and}$$

Differentiating with respect to time,

$$\frac{\partial \Psi}{\partial t} = \frac{-iE}{\hbar} \Psi \dots\dots\dots(3)$$

$$p^2 \Psi = -\hbar^2 \frac{\partial^2 \Psi}{\partial x^2} \dots\dots\dots(4)$$

$$= -i\hbar \frac{\partial}{\partial x} \left(-i\hbar \frac{\partial}{\partial x} \right) \Psi \dots\dots\dots(4a)$$

From 3 and 4, $p = -i\hbar \frac{\partial}{\partial x}$

This is called momentum operator:

$$E \Psi = i\hbar \frac{\partial \Psi}{\partial t} \dots\dots\dots(5)$$

$$\therefore E = i\hbar \frac{\partial}{\partial t} \dots\dots\dots(5a)$$

This is called Energy operator.

For a free particle total energy is given by

$$E = \frac{p^2}{2m}, \text{ since } V = 0$$

$$\text{ie } E \Psi = \frac{p^2}{2m} \Psi \dots\dots\dots(6)$$

Using equation (4) and (5), Equation (6) becomes

$$\frac{-\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} = i\hbar \frac{\partial \Psi}{\partial t} \dots\dots\dots(7)$$

This is Schrodinger's equation for a free particle in one dimension.

In three dimension it becomes for a free particle,

$$\frac{-\hbar^2}{2m} \nabla^2 \Psi = i\hbar \frac{\partial \Psi}{\partial t} \dots\dots\dots(8)$$

$$\text{where } \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

is called *Laplacian operator*

If the particle is moving under a potential $V(r,t)$ then equation (8) becomes

$$\left[\frac{-\hbar^2}{2m} \nabla^2 + V_{(r,t)} \right] \Psi = i\hbar \frac{\partial \Psi}{\partial t}$$

This is Schrodinger's time dependent equation

TIME INDEPENDENT SCHRÖDINGER EQUATION (STEADY STATE EQUATION)

According to Schrodinger, de- Broglie's wavelength holds good for any particle moving in any field of force with potential energy V.

Then total energy $E = \text{Kinetic energy} + \text{Potential Energy}$

$$E = \frac{1}{2} m v^2 + V$$

$$= \frac{p^2}{2m} + V$$

$$\text{ie } p^2 = 2m(E - V)$$

$$\text{or } p = [2m(E - V)]^{1/2}$$

The wave equation in Cartesian co-ordinate system can be written as

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} = \frac{1}{u^2} \frac{\partial^2 \Psi}{\partial t^2} \dots\dots\dots(9)$$

$$\text{where } \Psi = \Psi_0 e^{-i\omega t} \dots\dots\dots(10)$$

'u' is the velocity of motion of wave and $\Psi(x, y, z, t)$ represents the amplitude of the wave associated with the particle

From (10), we get

$$\frac{\partial^2 \Psi}{\partial t^2} = -\omega^2 \Psi$$

So Equation (9) becomes

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} = \frac{-\omega^2}{u^2} \Psi \dots\dots\dots(11)$$

$$\frac{\omega^2}{u^2} = \frac{(2\pi\nu)^2}{(\nu\lambda)^2} = \frac{4\pi^2}{\lambda^2}$$

So we have

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} = \frac{-4\pi^2}{\lambda^2} \Psi \quad \text{or}$$

$$\nabla^2 \Psi + \frac{4\pi^2}{\lambda^2} \Psi = 0 \dots\dots\dots(12)$$

Equation (12) is a general equation which is independent of time.

Let us now introduce the concept of de-Broglie wavelength.

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m(E - V)}}$$

$$\lambda^2 = \frac{h^2}{2m(E - V)}$$

Substituting the wavelength in eqn (12)

$$\nabla^2 \Psi + \frac{4\pi^2}{h^2} [2m(E - V)] \Psi = 0$$

or

$$\nabla^2 \Psi + \frac{8\pi^2 m}{h^2} (E - V) \Psi = 0$$

or

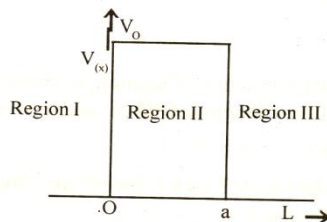
$$\nabla^2 \Psi + \frac{2m}{\hbar^2} (E - V) \Psi = 0 \dots \dots \dots (13)$$

This represents Schrodinger's time - independent wave equation.

For the case of a free particle (V=0), the Schrodinger equation becomes

$$\nabla^2 \Psi + \frac{2mE}{\hbar^2} \Psi = 0$$

QUANTUM MECHANICAL TUNELLING



Consider particle of mass m incident on the barrier from left

Classically

If $E < V_0$ all the particle will be reflected at the barrier

If $E > V_0$ all will be transmitted in to the region III

Quantum mechanically

If $E < V_0$ there will be a finite probability for the particle to be transmitted in to the region III

This phenomenon of tunnelling through the barrier is known as quantum mechanical tunnelling or barrier penetration . The solution of the schrodinger equation for the particle gives the wave function for the three region

Region I

wave function corresponds to two wave - one wave for the incident particle & other is for the reflected particle by the barrier

Region II

two solution are possible – an exponentially increasing function & an exponentially decreasing function

Region III

wave travelling from left to right

The concept is used to explain a number of phenomena in physics

1. The emission of α particles from the radioactive nuclei
2. Barrier penetration in tunnel diode and Josephson junction
3. Electron tunnelling in scanning tunnelling microscope

ONE DIMENSIONAL INFINITE SQUARE WELL POTENTIAL

Consider the motion of a particle of mass m confined to move in a one dimensional square well of infinite depth.

Since there is no interaction between the particle and wall, the potential energy of the particle is taken to be zero. It is very clear that potential energy does not depend on time and we are considering only time independent Schrodinger equation for the solution. The problem is one dimensional and the Schrodinger equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V) \psi = 0$$

since $V = 0$, the equation becomes

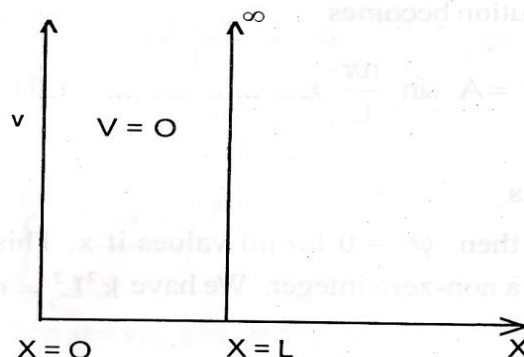
$$\frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2} \psi = 0 \quad \text{or}$$

$$\frac{d^2\psi}{dx^2} + k^2 \psi = 0 \quad \text{where } k^2 = \frac{2mE}{\hbar^2} \dots\dots\dots(17)$$

The solution of the above differential equation is of the form

$$\psi = A \sin kx + B \cos kx \dots\dots\dots(18)$$

The solution has to be well behaved wave function. Since the particle is inside a square well of infinite depth, it is impossible to find the particle outside. i.e. ψ must be zero for all points outside the square well.



$$\psi = 0 \text{ for } x < 0 \text{ and}$$

$$\psi = 0 \text{ for } x > L$$

This is possible only if $\psi = 0$ at $x = 0$ and $x = L$ as demanded by continuity condition.

Applying first condition on equation 18, we get

$$0 = A \sin 0 + B \cos 0$$

$$\text{ie } B = 0$$

So solution reduces to

$$\psi = A \sin kx \dots \dots \dots (19)$$

Using the next condition $\psi = 0$ at $x = L$, we get

$$0 = A \sin kL$$

There are two possibilities. Either $A = 0$ or $\sin kL = 0$

A cannot be zero, since the wavefunction cannot exist. So we have the other possibility

$$\sin kL = 0$$

$$\text{ie } kL = n\pi \dots \dots \dots (20)$$

where n is an integer.

So the solution becomes

$$\psi = A \sin \frac{n\pi}{L} x \dots \dots \dots (21)$$

Eigen Values

If $n = 0$, then $\psi = 0$ for all values of x . This is ruled out.

Therefore n is a non-zero integer. We have $k^2 L^2 = n^2 \pi^2$ on using equation (17)

$$k^2 = 2mE/\hbar^2$$

$$\frac{2mE}{\hbar^2} L^2 = n^2 \pi^2$$

$$E = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \quad \text{or}$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \quad \text{ie} \quad E_n = \frac{n^2 h^2}{8mL^2}$$

E_n as the energy corresponding to n . The quantity n is called quantum number. The different values of energy for n are called **Eigen values**. Since n is restricted, the particle cannot have any value of energy but will be restricted to certain values.

Eigen wave functions

The wave functions associated with different Eigen values of energy are called **Eigen Functions**.

By applying the normalization condition,

$$\int \psi^* \psi dx dy dz = 1$$

So we have $\int_0^L A \sin \frac{n\pi x}{L} A \sin \frac{n\pi x}{L} dx = 1$

$$A^2 \int_0^L \sin^2 \frac{n\pi x}{L} dx = 1$$

$$A^2 \int_0^L \frac{1 - \cos 2 \left(\frac{n\pi x}{L} \right)}{2} dx = 1.$$

On integration and applying limits we get

$$A^2 \frac{L}{2} = 1 \quad \text{or } A = \sqrt{\frac{2}{L}}$$

So the normalised wave function is

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad \text{where } n = 1, 2, 3, \dots \quad (22)$$

For the ground state $n=1$, the wave function is given by

$$\psi_1 = \sqrt{\frac{2}{L}} \sin \frac{\pi x}{L}$$

Similarly the wave function for the first two excited states are given by

$$\psi_2 = \sqrt{\frac{2}{L}} \sin \frac{2\pi x}{L} \quad \psi_3 = \sqrt{\frac{2}{L}} \sin \frac{3\pi x}{L}$$

These wave functions associated with different energy are called Eigen wave functions.

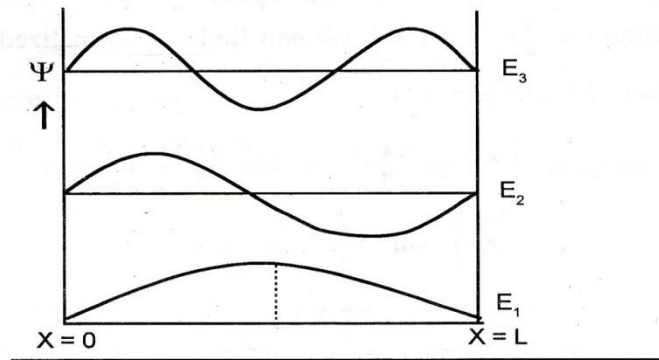


Fig. 2 wave function of first three energy levels

